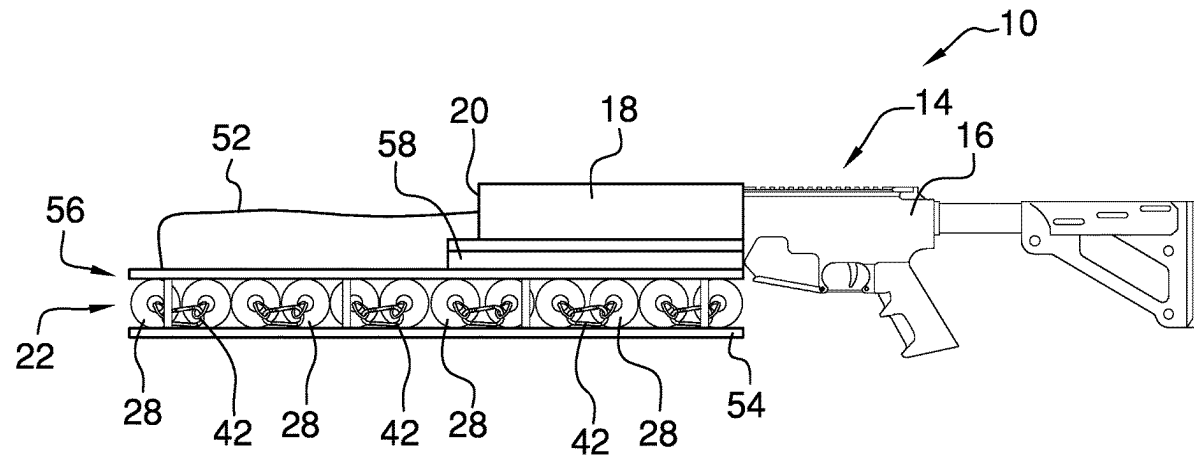


(12) **Patent Application Publication**
Tanner

(43) **Pub. Date:** **Sep. 11, 2025**



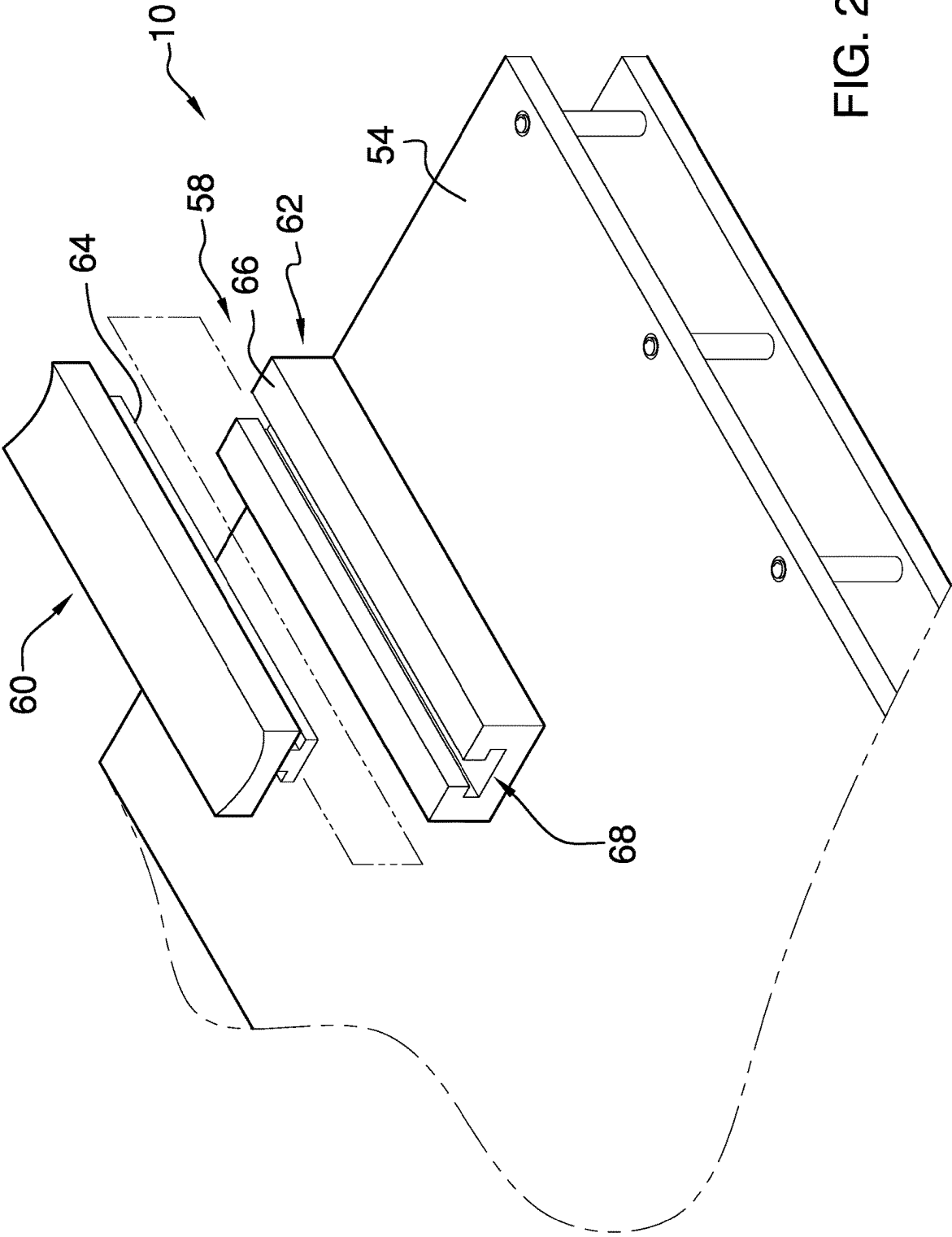
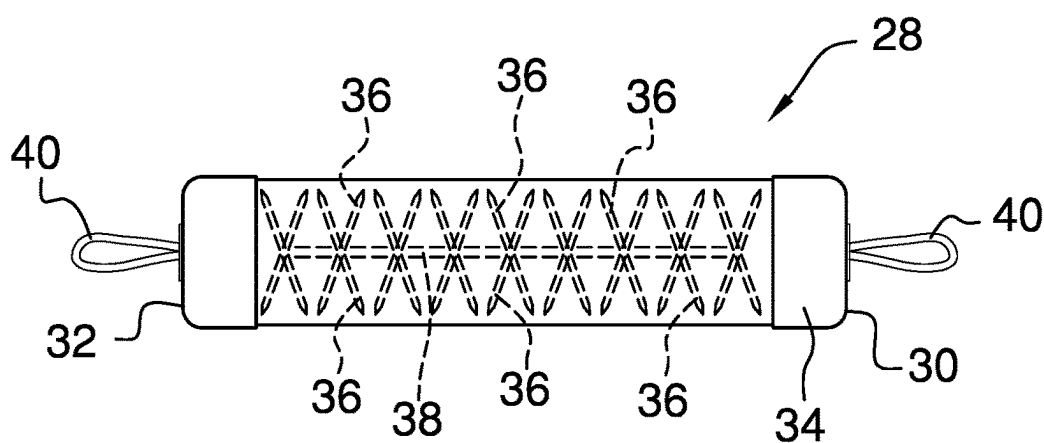
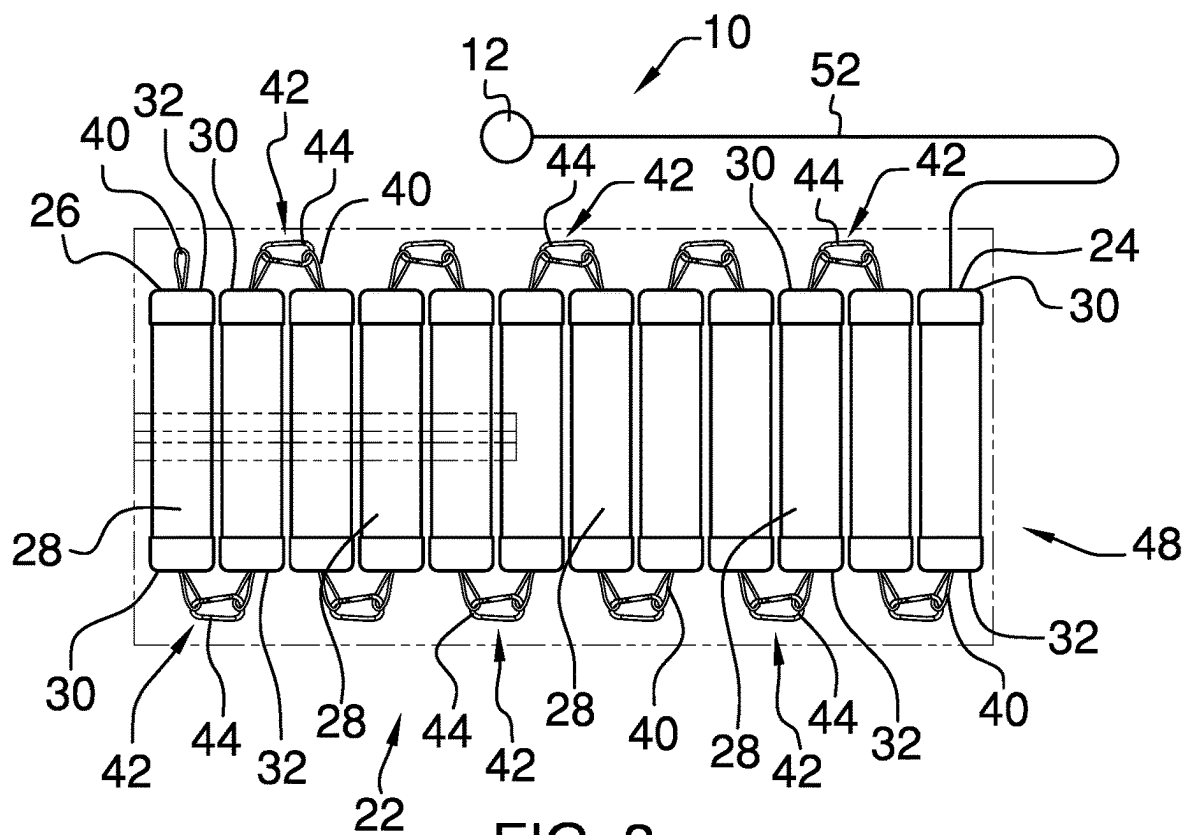
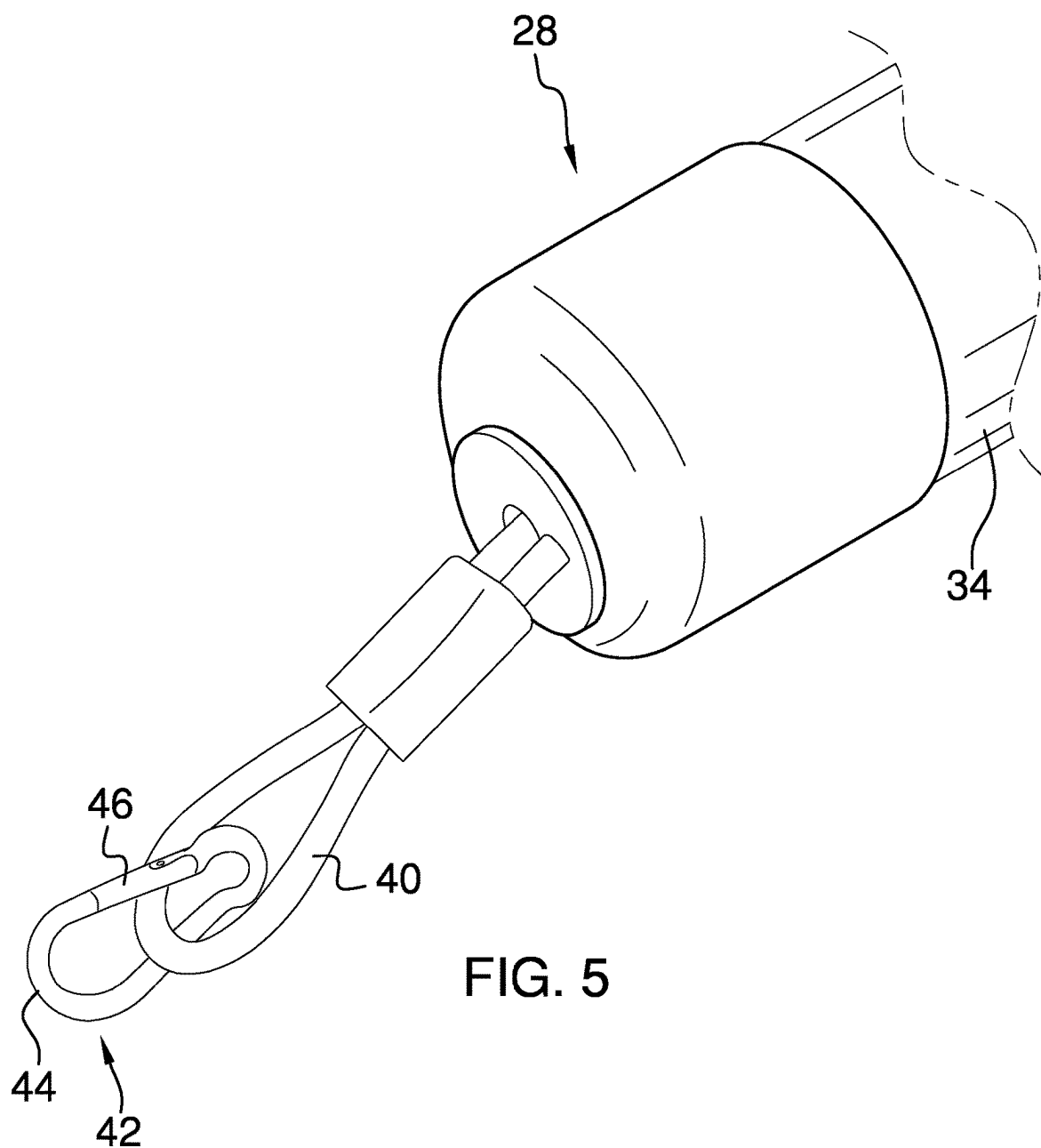


FIG. 2





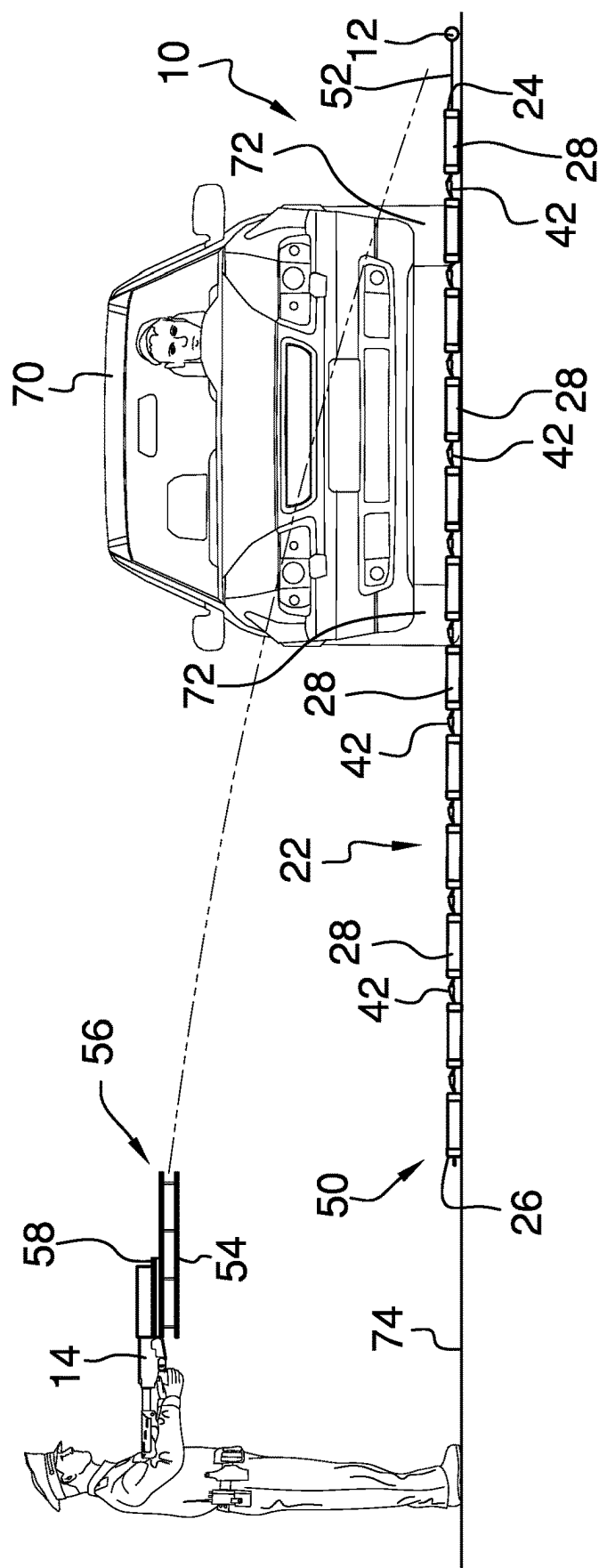


FIG. 6

VEHICLE ARRESTING APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] Not Applicable

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not Applicable

THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

[0003] Not Applicable

INCORPORATION-BY-REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISC OR AS A TEXT FILE VIA THE OFFICE ELECTRONIC FILING SYSTEM

[0004] Not Applicable

STATEMENT REGARDING PRIOR DISCLOSURES BY THE INVENTOR OR JOINT INVENTOR

[0005] Not Applicable

BACKGROUND OF THE INVENTION**(1) Field of the Invention**

[0006] The disclosure relates to vehicle arresting apparatuses and more particularly pertains to a new vehicle arresting apparatus for launching a puncturing device onto a driving surface to puncture a motor vehicle's tires.

(2) Description of Related Art Including Information Disclosed Under 37 CFR 1.97 and 1.98

[0007] The prior art describes myriad vehicle arresting devices, including devices which deploy a piercing member across a road surface from one side in order to deflate a motor vehicle's tires. U.S. Pat. No. 10,030,343, for example, discloses a launching device which casts a flexible net with attached spikes across a road surface. The prior art fails, however, to disclose an apparatus which launches a projectile from a projectile launcher that draws a deflation member comprised of alternating rigid spiked members and flexible connectors across a driving surface.

BRIEF SUMMARY OF THE INVENTION

[0008] An embodiment of the disclosure meets the needs presented above by generally comprising a projectile and a projectile launcher configured to launch the projectile. A deflation member is coupled to the projectile. The deflation member comprises a plurality of puncture members and a plurality of flexible links which are coupled together in alternating fashion. Each puncture member is configured to puncture a tire, and each flexible link couples a first end of one of the puncture members to a second end of another of the puncture members.

[0009] There has thus been outlined, rather broadly, the more important features of the disclosure in order that the detailed description thereof that follows may be better

understood, and in order that the present contribution to the art may be better appreciated. There are additional features of the disclosure that will be described hereinafter and which will form the subject matter of the claims appended hereto.

[0010] The objects of the disclosure, along with the various features of novelty which characterize the disclosure, are pointed out with particularity in the claims annexed to and forming a part of this disclosure.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWING(S)

[0011] The disclosure will be better understood and objects other than those set forth above will become apparent when consideration is given to the following detailed description thereof. Such description makes reference to the annexed drawings wherein:

[0012] FIG. 1 is a side view of a vehicle arresting apparatus according to an embodiment of the disclosure.

[0013] FIG. 2 is a perspective detail view of a cartridge and a cartridge mount of an embodiment of the disclosure.

[0014] FIG. 3 is a detail view of a deflation member of an embodiment of the disclosure in a stored configuration.

[0015] FIG. 4 is a detail view of a puncture member of a deflation member of an embodiment of the disclosure.

[0016] FIG. 5 is a detail view of a flexible link and a puncture member of a deflation member of an embodiment of the disclosure.

[0017] FIG. 6 is an in-use view of an embodiment of the disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0018] With reference now to the drawings, and in particular to FIGS. 1 through 6 thereof, a new vehicle arresting apparatus embodying the principles and concepts of an embodiment of the disclosure and generally designated by the reference numeral 10 will be described.

[0019] As best illustrated in FIGS. 1 through 6, the vehicle arresting apparatus 10 generally comprises a projectile 12 and a projectile launcher 14 configured to launch the projectile 12. The projectile launcher 14 comprises a body 16 and a barrel 18. The barrel 18 is coupled to and extends forwardly from the body 16 to a front end 20 which is open to facilitate release of the projectile 12 through the front end 20. The projectile 12 is receivable into the barrel 18 and launchable out of the barrel 18 upon actuation of a firing mechanism of the projectile launcher 14. The firing mechanism of the projectile launcher 14 may use explosive charges, compressed gas, linear induction motors, or any other suitable propellant for launching the projectile 12 from the projectile launcher 14. In some cases, explosive charges or other energy stores may be coupled to the projectile 12 prior to launching the projectile 12.

[0020] A deflation member 22 is coupled to the projectile 12 via a tether attached to a leading end 24 of the deflation member 22. The deflation member 22 comprises a plurality of puncture members 28 and a plurality of flexible links 42 which are coupled together in alternating fashion. It is understood that the term "flexible link" is used in the present disclosure to refer to flexibility afforded the deflation member 22 which allows the deflation member to bend at the location of the flexible link. The flexible link may in fact

comprise a rigid material such that the flexible link itself does not bend or otherwise deform.

[0021] The puncture members 28 are substantially rigid such that they maintain their shapes prior to use in puncturing a tire 72 of a motor vehicle 70. Each flexible link 42 couples a first end 30 of one of the puncture members 28 to a second end 32 of another of the puncture members 28. The puncture members 28 are elongated between the first and second ends 30, 32 such that the deflation member 22 may be placed into a linear configuration 50 in which the puncture members 28 align with each other along a central axis of the deflation member 22.

[0022] Each puncture member 28 comprises a capsule 34 and a plurality of quills 36. The quills 36 are housed by the capsule 34 and are attached to each other by a rod 38 or flexible line that extends between the first and second ends 30, 32 of the puncture member 28. The capsule 34 comprises a material with a strength sufficient to rigidly maintain its shape until a weight force of the motor vehicle 70 is applied, at which point the capsule 34 splits and exposes one or more quills 36 from within the capsule 34. The material of the capsules 34 may be a rigid plastic or a ceramic which fractures under the weight force, a flexible plastic or the like, which collapses and allows one or more quills 36 to pierce the capsule 34, or any other suitable material. Each quill 36 has pointed ends configured to pierce one of the tires 72 of the motor vehicle 70 and is hollow. When any of the quills 36 becomes embedded in the tire 72, the quill 36 facilitates passage of air from the tire 72 to a surrounding atmosphere.

[0023] Each flexible link 42 comprises a clip 44 with a gate 46 biased to form a closed loop. Each puncture member 28 comprises at least one attachment loop 40 which is interlocked with one of the flexible links 42. The clip 44 may be a carabiner, or the flexible link 42 may comprise other suitable connection means such as a quick link chain link, a cable, or the like. The clip 44 may be opened to separate the adjacent puncture members 28 from each other for removal, replacement, or repair. Other embodiments also may have flexible links 42 which do not readily detach from the adjacent puncture members 28. Most of the puncture members 28 will have two attachment loops 40 for attaching to flexible links 42 positioned on either of the first and second ends 30, 32 of the puncture member 28. But the puncture members 28 at the leading end 24 and a trailing end 26 of the deflation member 22 may only have one attachment loop 40 for attaching to an adjacent one of the flexible links 42.

[0024] A cartridge 54 is mounted to the projectile launcher 14 which contains the deflation member 22 in a stored configuration 48. The deflation member 22 follows a serpentine path when in the stored configuration 48 wherein the puncture members 28 are arranged along a path substantially parallel to the barrel 18. The puncture members 28 also lie parallel to each other and perpendicular to the path when in the stored configuration 48. The cartridge 54 has an open forward end 56 such that the deflation member 22 may be removed forwardly from the cartridge 54. The tether 52 extends through the open forward end 56 of the cartridge 54 when the deflation member 22 is in the stored configuration 48 so the projectile 12 may be positioned within the barrel 18 of the projectile launcher 14 when the cartridge 54 stores the deflation member 22.

[0025] A cartridge mount 58 removably couples the cartridge 54 to the barrel 18 of the projectile launcher 14. The cartridge mount 58 comprises a first mating member 60 coupled to the projectile launcher 14 and a second mating member 62 coupled to the cartridge 54. The first mating member 60 comprises a rail 64, and the second mating member 62 comprises a slider 66. The rail 64 is oriented parallel to the barrel 18 and has a T-shaped cross section. The slider 66 defines a cavity 68 complementary in shape to the rail 64. Threaded fasteners, locking pins, or the like may be used to secure the slider 66 to the rail 64. In other embodiments, the cartridge mount 58 may comprise another suitable fastener including a bracket and threaded fasteners, a clamp, a snap connection, or the like.

[0026] In use, the cartridge 54 is coupled to the projectile launcher 14 via the cartridge mount 58 while containing the deflation member 22 in the stored configuration 48. The projectile 12 is placed in the barrel 18 of the projectile launcher 14. To deploy the deflation member 22 across a driving surface 74, the firing mechanism of the projectile launcher 14 is actuated to launch the projectile 12, which draws the deflation member 22 via the leading end 24 away from the projectile launcher 14. As the deflation member 22 is drawn out of the cartridge 54, each puncture member 28 is pulled into alignment with the other puncture members 28 and generally parallel to the direction of travel of the projectile 12. The deflation member 22 is thereby positioned in the linear configuration 50 by nature of the operation of the projectile launcher 14 and falls to rest on the driving surface 74.

[0027] When the motor vehicle 70 drives over one or more of the puncture members 28 of the deflation member 22, the capsules 34 bearing the weight of the motor vehicle 70 split, and the quills 36 therein puncture the tires 72 of the motor vehicle 70. The quills 36 become embedded in the tires 72, which allows air to be forced out of the tires 72 via the pressure differential between the internal chambers of the tires 72 and the surrounding atmosphere. By deflating the tires 72, the motor vehicle 70 loses traction with the driving surface 74 and is thus immobilized or decelerated by the vehicle arresting apparatus 10.

[0028] With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of an embodiment enabled by the disclosure, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by an embodiment of the disclosure.

[0029] Therefore, the foregoing is considered as illustrative only of the principles of the disclosure. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the disclosure to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the disclosure. In this patent document, the word "comprising" is used in its non-limiting sense to mean that items following the word are included, but items not specifically mentioned are not excluded. A reference to an element by the indefinite article "a" does not exclude the possibility that more than one of the element is present, unless the context clearly requires that there be only one of the elements.

I claim:

1. A vehicle arresting apparatus comprising:
 - a projectile;
 - a projectile launcher configured to launch the projectile; and
 - a deflation member coupled to the projectile, the deflation member comprising a plurality of puncture members and a plurality of flexible links, the puncture members and the flexible links being coupled together in alternating fashion, each flexible link coupling a first end of one of the puncture members to a second end of another of the puncture members, each puncture member being configured to puncture a tire.
2. The apparatus of claim 1, wherein the projectile launcher comprises a body and a barrel, the barrel being coupled to and extending forwardly from the body, the barrel being open at a front end of the barrel, the projectile being receivable into the barrel and launchable out of the barrel upon actuation of the projectile launcher.
3. The apparatus of claim 2, further comprising a cartridge mounted to the projectile launcher, the cartridge containing the deflation member in a stored configuration, the deflation member being removable forwardly from the cartridge upon launching the projectile via the projectile launcher.
4. The apparatus of claim 3, the deflation member following a serpentine path when in the stored configuration wherein the puncture members lie along a path substantially parallel to the barrel.
5. The apparatus of claim 3, further comprising a tether coupling the deflation member to the projectile, the tether being attached to a leading end of the deflation member, the tether extending through an open forward end of the cartridge when the deflation member is in the stored configuration.
6. The apparatus of claim 3, further comprising a cartridge mount removably coupling the cartridge to the barrel of the projectile launcher, the cartridge mount comprising a first mating member coupled to the projectile launcher and a second mating member coupled to the cartridge.
7. The apparatus of claim 6, wherein the first mating member comprises a rail, the second mating member comprising a slider, the rail being oriented parallel to the barrel.
8. The apparatus of claim 7, wherein the rail has a T-shaped cross section, the slider defining a cavity complementary in shape to the rail.
9. The apparatus of claim 1, wherein each puncture member is substantially rigid.
10. The apparatus of claim 1, wherein each puncture member comprises a capsule and a plurality of quills, the plurality of quills being housed by the capsule, the capsule being configured to split under a weight of a motor vehicle, each quill having pointed ends configured to pierce the tire.
11. The apparatus of claim 10, wherein each quill is hollow wherein each quill facilitates a passage of air from the tire to an atmosphere when the quill is embedded in the tire.
12. The apparatus of claim 1, wherein each flexible link comprises a clip with a gate biased to form a closed loop.
13. The apparatus of claim 1, wherein each puncture member comprises an attachment loop which is interlocked with one of the flexible links.
14. The apparatus of claim 1, further comprising a tether coupling a leading end of the deflation member to the projectile.
15. The apparatus of claim 1, further comprising a cartridge mounted to the projectile launcher, the cartridge containing the deflation member in a stored configuration, the cartridge having an open forward end such that the deflation member may be removed through the open forward end.
16. The apparatus of claim 15, wherein the deflation member follows a serpentine path when in the stored configuration.
17. The apparatus of claim 16, wherein the puncture members are elongated and lie parallel to each other when the deflation member is in the stored configuration.
18. The apparatus of claim 15, further comprising a cartridge mount removably coupling the cartridge to the projectile launcher, the cartridge mount comprising a first mating member coupled to the projectile launcher and a second mating member coupled to the cartridge.
19. The apparatus of claim 18, wherein the first mating member comprises a rail, the second mating member comprising a slider, the rail having a T-shaped cross section, the slider defining a cavity complementary in shape to the rail.
20. A vehicle arresting apparatus comprising:
 - a projectile;
 - a projectile launcher configured to launch the projectile, the projectile launcher comprising a body and a barrel, the barrel being coupled to and extending forwardly from the body, the barrel being open at a front end of the barrel, the projectile being receivable into the barrel and launchable out of the barrel upon actuation of the projectile launcher;
 - a deflation member coupled to the projectile, the deflation member comprising a plurality of puncture members and a plurality of flexible links, the puncture members and the flexible links being coupled together in alternating fashion, each flexible link coupling a first end of one of the puncture members to a second end of another of the puncture members, each puncture member being substantially rigid, each puncture member comprising a capsule and a plurality of quills, the plurality of quills being housed by the capsule, the capsule being configured to split under a weight of a motor vehicle, each quill having pointed ends configured to pierce a tire, each quill being hollow wherein each quill facilitates a passage of air from the tire to an atmosphere when the quill is embedded in the tire, each flexible link comprising a clip with a gate biased to form a closed loop, each puncture member comprising an attachment loop which is interlocked with one of the flexible links;
 - a tether coupling a leading end of the deflation member to the projectile;
 - a cartridge mounted to the projectile launcher, the cartridge containing the deflation member in a stored configuration, the deflation member following a serpentine path when in the stored configuration wherein the puncture members lie along a path substantially parallel to the barrel, the cartridge having an open forward end such that the deflation member may be removed through the open forward end, the tether extending through the open forward end of the cartridge when the deflation member is in the stored configuration, the puncture members being elongated and lying parallel to each other when the deflation member is in the stored configuration; and

a cartridge mount removably coupling the cartridge to the barrel of the projectile launcher, the cartridge mount comprising a first mating member coupled to the projectile launcher and a second mating member coupled to the cartridge, the first mating member comprising a rail, the second mating member comprising a slider, the rail being oriented parallel to the barrel, the rail having a T-shaped cross section, the slider defining a cavity complementary in shape to the rail.

* * * * *